Dedeepya Majety

Azure Data Engineer & Lakehouse Specialist | Data Engineer

Accenture in India | Data & AI Platform Engineering | Req ID: 4476918574

✔ Associate - Azure Data Engineer @ Cognizant (3.9 YOE delivering enterprise cloud lakehouse solutions).
✔ Databricks Certified Data Engineer Associate (ID: 185421754) | Microsoft Power BI Associate (PL-300).
✔ Verified Track Record: Modernized 40+ legacy Informatica workflows to Databricks Delta Lake (40% runtime reduction).
✔ 🟢 Serving Official Notice Period at Cognizant | Available to Join in 30 Days (Immediate Joiner Advantage).

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My Technical Strategy for Accenture in India: 3 Core Pillars

High-throughput data engineering, cost efficiency, and automated governance

1. Lakehouse Modernization

Modernization

- Migrate legacy batch ETL to Azure Databricks Delta Lake and PySpark.
- Implement continuous Auto Loader (cloudFiles) for streaming file ingestion.
- Ensure 100% data parity with automated row-count and checksum validation.
- Cut pipeline runtimes by ~40% and cloud compute spend by ~30%.

2. Performance & Scale

Scale & Speed

- Mitigate distributed data skew using key salting and broadcast hash joins.
- Leverage Adaptive Query Execution (AQE) for dynamic shuffle partition tuning.
- Compact small files via OPTIMIZE and Z-ORDER on high-cardinality keys.
- Orchestrate modular Python Airflow DAGs with 99.8% SLA adherence.

3. Governance & Quality

Data Trust

- Enforce centralized Unity Catalog 3-level namespace (catalog.schema.table).
- Implement dynamic column masking and row-level security filters (RBAC).
- Tune Informatica MDM match-merge trust scores for unified golden records.
- Build executive Power BI dashboards with optimized DAX models.

Production Medallion Lakehouse Architecture

End-to-end multi-hop data refinement, ACID reliability, and analytics serving

Bronze Tier: Raw Ingestion

Ingestion Tier

- Source: ADLS Gen2 object storage with multi-source file drops (JSON, CSV, Parquet).
- Engine: Databricks Auto Loader (cloudFiles) with RocksDB state storage.
- Schema Evolution: Automated schema inference with _rescued_data column protection.
- Storage: Append-only Delta Lake raw tables preserving source fidelity.
- SLA: Near-real-time streaming and scheduled micro-batch extraction.

Silver Tier: Conformed & Clean

Curated Tier

- Processing: PySpark distributed cleaning and business rule validation.
- Deduplication: Window functions (row_number() over primary keys).
- Idempotency: Delta Lake ACID MERGE INTO with SCD Type 1 & Type 2 tracking.
- Governance: Unity Catalog table/column access controls and lineage.
- Quality: Automated schema enforcement and data contract assertions.

Gold Tier: Business Data Marts

Serving Tier

- Data Modeling: Curated Star Schema dimensional models (Fact & Dim tables).
- Optimization: File compaction via OPTIMIZE and Z-ORDER on query dimensions.
- Serving: High-concurrency Databricks SQL Serverless Warehouses.
- BI Layer: Power BI Direct Lake connectivity with optimized DAX measures.
- Impact: 25% query latency improvement across operational scorecards.

Enterprise Production Track Record (Cognizant / Amgen)

Verified technical levers and quantitative delivery impact across 3.9 YOE

40% Pipeline Runtime Reduction

40% Faster

- Migrated 40+ legacy Informatica PowerCenter workflows to Azure Databricks.
- Transitioned fixed-schedule batch extracts to Auto Loader streaming.
- Slashed daily execution windows from 5 hours to under 3 hours.

~30% Cloud Infrastructure Savings

30% Savings

- Rightsized Azure Databricks cluster nodes and enabled auto-termination.
- Leveraged spot compute instances for non-critical batch transformation stages.
- Eliminated legacy on-premises license dependencies and maintenance overhead.

99.8% Pipeline SLA Reliability

99.8% Uptime

- Authored modular Python Apache Airflow DAGs with exponential backoff retries.
- Configured automated alerting and automated failure isolation across ADLS Gen2.
- Guaranteed 100% data parity between legacy feeds and Delta Lake targets.

~15% Duplicate Entities Eliminated

Golden Records

- Tuned Informatica MDM match-and-merge rule sets and survivorship logic.
- Resolved entity duplication across global customer and organization master records.
- Delivered 6+ executive Power BI dashboards with custom DAX scorecards.

PySpark Distributed Tuning & Troubleshooting Mastery

Solving distributed data skew, memory bottlenecks, and executor failures at the root

Data Skew Diagnosis & Salting

Skew Fix

- Symptom: 199 tasks finish in seconds; 200th task runs for 30 mins (spill to disk).
- Root Cause: Hotspot keys (e.g. nulls or massive enterprise IDs) piling on one executor.
- Fix: Salting keys via concat(key, floor(rand() * 10)) to distribute partitions uniformly.

Broadcast Hash Join Optimization

Zero Shuffle

- Problem: Expensive SortMergeJoin triggering multi-gigabyte network shuffle stages.
- Fix: Force broadcast(dim_df) for dimension tables under 10MB.
- Outcome: Eliminates network shuffle entirely, cutting join latency by up to 70%.

Memory Management & 137 OOMs

OOM Shield

- Symptom: java.lang.OutOfMemoryError or ExecutorLostFailure (exit code 137).
- Root Causes: Uncontrolled collect(), driver overwhelm, or memory/storage contention.
- Fix: Replace collect() with take()/write, tune spark.memory.fraction, and persist properly.

Delta Compaction & Small File Swamps

Compaction

- Symptom: Streaming sink writes 10,000 sub-MB files, stalling downstream query planning.
- Fix: Configure spark.databricks.delta.autoCompact.enabled = true and optimizeWrite.
- Maintenance: Schedule hourly OPTIMIZE table ZORDER BY (keys) + VACUUM.

Enterprise Governance & DataOps Architecture

End-to-end data security, regulatory compliance, and golden master record integrity

Databricks Unity Catalog Governance

Unity Catalog

- Three-Level Namespace: Clean hierarchical governance (catalog.schema.table).
- Role-Based Access Control (RBAC): Fine-grained permissions across multi-tenant environments.
- Column Masking: Dynamic redaction of PII / sensitive attributes based on user group membership.
- Row-Level Filtering: Restricting data access based on geographical or business entity scope.
- End-to-End Data Lineage: Automated tracking from raw source files to analytical dashboards.
- Audit Compliance: Full visibility into table alterations and user query histories.

Informatica MDM & Data Quality

Master Data

- Match-and-Merge Rules: Configured deterministic exact match and probabilistic fuzzy match logic.
- Trust Decay Scoring: Established attribute-level trust curves favoring verified modern CRM inputs.
- Survivorship Logic: Automated resolution of conflicting source values to create unified golden records.
- Entity Deduplication: Reduced master customer and organization duplicate rates by ~15%.
- Airflow DataOps: Automated data quality gates verifying row-counts and checksums before publish.
- GxP Awareness: Rigorous adherence to audit trails and regulatory data validation requirements.

My Operating Principles: Velocity & High-Agency Ownership

Core professional standards defining day-to-day engineering execution

1. Mathematical Parity Before Cutover

Zero Loss

- Never migrate an ETL pipeline based on gut feeling or spot checks.
- Enforce automated row-count, primary key nullness, and checksum reconciliation.
- Guarantee 100% data parity between legacy systems and modern Lakehouse sinks.

2. Proactive Schema Drift Defense

Resilience

- External source schemas will inevitably change unexpectedly at midnight.
- Build resilient ingestion using Auto Loader's _rescued_data to avoid batch halts.
- Implement schema contracts and automated alerting to resolve drift before business impact.

3. Root Cause Performance Tuning

Efficiency

- Never solve a Spark performance bottleneck by blindly doubling cluster compute size.
- Inspect Spark UI task timelines to isolate the exact stage causing data skew or memory spill.
- Apply architectural fixes (salting, broadcasting, partition pruning) to permanently resolve issues.

4. Full-Lifecycle Engineering Ownership

Ownership

- Take complete ownership from raw ingestion to the final executive dashboard scorecard.
- Collaborate seamlessly across data architecture, business operations, and QA teams.
- Document data lineage, design decisions, and runbooks for transparent team velocity.

30-60-90 Day Execution Roadmap for Accenture in India

Phased delivery plan ensuring immediate day-1 contribution and long-term technical value

Days 1 to 30: Immersion & Audit

Days 1-30

- Audit existing data pipelines, Airflow DAGs, and ADLS Gen2 storage tiers.
- Identify compute cost hotspots, query bottlenecks, and data skew risks.
- Engage with data leads and business stakeholders to map critical SLA dependencies.
- Take ownership of primary ETL batch monitoring and incident response.
- Deliver initial pipeline bug fixes with 100% verified test coverage.

Days 31 to 60: Modernize & Tune

Days 31-60

- Modernize priority batch workflows to Databricks Auto Loader and Delta Lake.
- Implement PySpark optimization patterns (salting, broadcasting, partition pruning).
- Configure automated compaction (OPTIMIZE/Z-ORDER) and VACUUM maintenance jobs.
- Establish automated data quality reconciliation gates in CI/CD pipelines.
- Achieve measurable runtime reductions across core analytical feeds.

Days 61 to 90: Governance & Scale

Days 61-90

- Expand Databricks Unity Catalog governance (RBAC, column masking, lineage).
- Design curated star-schema data marts for high-concurrency BI serving.
- Partner with BI analysts to optimize Direct Lake connectivity and DAX performance.
- Author comprehensive architecture documentation, runbooks, and SOPs.
- Mentor junior engineers on Delta Lake internals and PySpark optimization.